ALEXANDRE CARON

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RESEARCH INTERESTS

High-energy theoretical physics; quantum field theory in curved spacetime; general relativity and gravitational physics; mathematical physics.

EDUCATION

Bachelor of Science - Honours Mathematics and Physics Fall 2024 - Winter 2027
McGill University: CGPA 3.97/4.00

PAST ACADEMIC EXPERIENCE

Undergraduate Summer Project Summer 2026
McGill University – Department of Physics

I studied the Unruh and Hawking effects from two perspectives: first using the framework of quantum field theory in curved spacetime, and then using a lesser-known approach developed by Parikh and Wilczek that treats black hole radiation as a tunneling process. Building on this second formalism, I independently formulated a possible original derivation of the Unruh effect and investigated its physical consequences.

I was supervised by Prof. Keshav Dasgupta.

Responsibilities

- Read and worked through primary literature and textbook material on semiclassical gravity and QFT in curved spacetime, including Parikh & Wilczek (2000) and chapters 6-9 of Mukhanov & Winitzki's textbook *Introduction to Quantum Effects in Gravity*
- Presented progress and findings to my supervisor on a weekly basis
- Currently writing a review article on the subject (in progress)

Skills Developed

- Independent research and critical reading of primary literature in theoretical physics
- Formulating and pursuing an original research question with minimal guidance
- Scientific writing, through the review article currently in preparation
- Developed basic knowledge of general relativity (to the level of Carroll's *Spacetime and Geometry*) and scalar field theory (first two chapters of Peskin & Schroeder's *An Introduction to Quantum Field Theory*)

Directed Reading Program Winter 2026
McGill University – Department of Mathematics and Statistics

The project's aim was to understand how symmetries, in the form of Lie groups and algebras, are used in the study of differential equations. Methods to solve exactly large classes of equations, and the theory of solvability of ODEs, were investigated. At the end of the semester, an introductory paper covering the subject was written, and a talk on the same subject was presented to the Department of Mathematics and Statistics.

I was supervised by Ph. D. student Shereen Elaidi, and worked in collaboration with fellow undergraduate Xavier Zurowski.

Responsibilities

- Studied Lie group and Lie algebra methods for solving ODEs through directed reading of textbooks, including Hydon's *Symmetry Methods for Differential Equations* and the first two chapters of Olver's *Applications of Lie Groups to Differential Equations*

- Co-wrote an introductory paper explaining the subject to a non-specialist audience
- Presented the material in a talk to the Department of Mathematics and Statistics

Skills Developed

- Self-directed learning of advanced mathematical material
- Technical writing aimed at a non-specialist audience
- Oral presentation of mathematical content

Physics Tutor Winter 2024
Cégep de Saint-Hyacinthe

I held three one-on-one sessions with students per week.

Responsibilities

- Reviewed concepts seen in class
- Worked through problems with students, including reviewing missed exam questions
- Gave general problem-solving strategies and study techniques

Skills Developed

- A deeper understanding of the subject, gained by having my own grasp of it challenged
- Stronger communication skills

AWARDS

Laureate of the AMQ Math Competition Winter 2024

The AMQ (Association Mathématique du Québec) holds a 3-hour annual competition in most cégeps in Quebec. I achieved one of the best scores in the province, placing 24th overall. A list of laureates and past competitions is available on their website: `https://www.amq.math.ca/concours/concours-precedents/`.

TALKS

Directed Reading Program – End of Term Talk Winter 2026

An introductory talk on Lie symmetries and the solvability of ordinary differential equations, presented to the McGill Department of Mathematics and Statistics.

SKILLS AND HOBBIES

Programming

Python: comfortable with NumPy, Matplotlib, Pandas, and SciPy.

C++: developed simple 2D simulations using SFML, including n gravitationally-interacting bodies with collisions, and systems of thousands of non-interacting particles in a central potential. Also solved problems on *Project Euler* (`https://projecteuler.net/`), a library of math/computer science challenges.

Music

I have played piano for around 10 years and enjoy improvising in my free time. I taught myself composition and write my own pieces using MuseScore.